EthosLogAnalyser – User guide

For pilots who use the Windows EXE. Version: 1.0.0. No programming knowledge needed. German version: user-guide_de.md.

EthosLogAnalyser reads the flight logs (CSV files) that your **FrSky Ethos** transmitter writes and turns them into a **logbook** of your flights, **charts** of every flight, a **flight recorder** that replays a flight, and **statistics** about your batteries and your model. All data stays on your computer; your log files are never changed.

The screenshots show the English interface. The program also speaks German and Spanish (see Language).

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1. Install and start

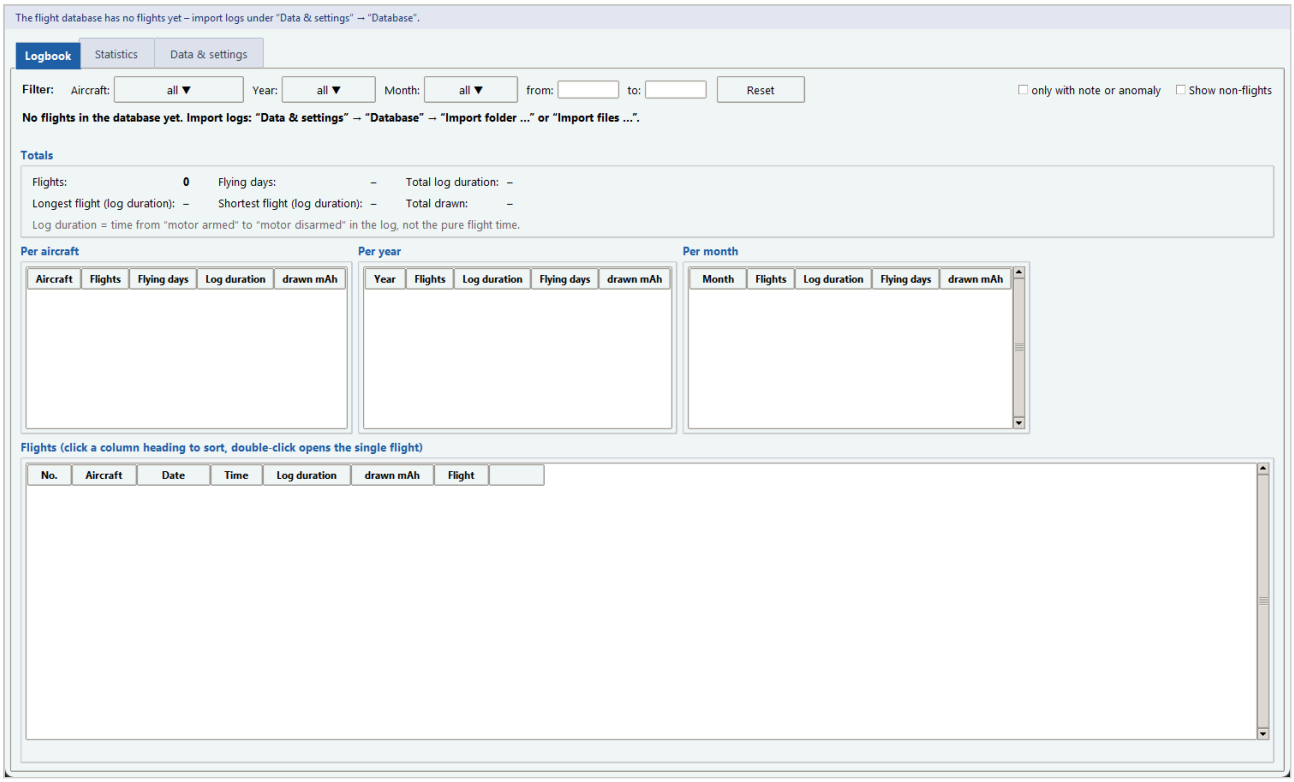
There is no installer. Download the `.exe` of the newest release from the Releases page, put it in any folder and double-click it.

Windows may show a blue **SmartScreen** warning because the file is not signed: click **More info** → **Run anyway**.

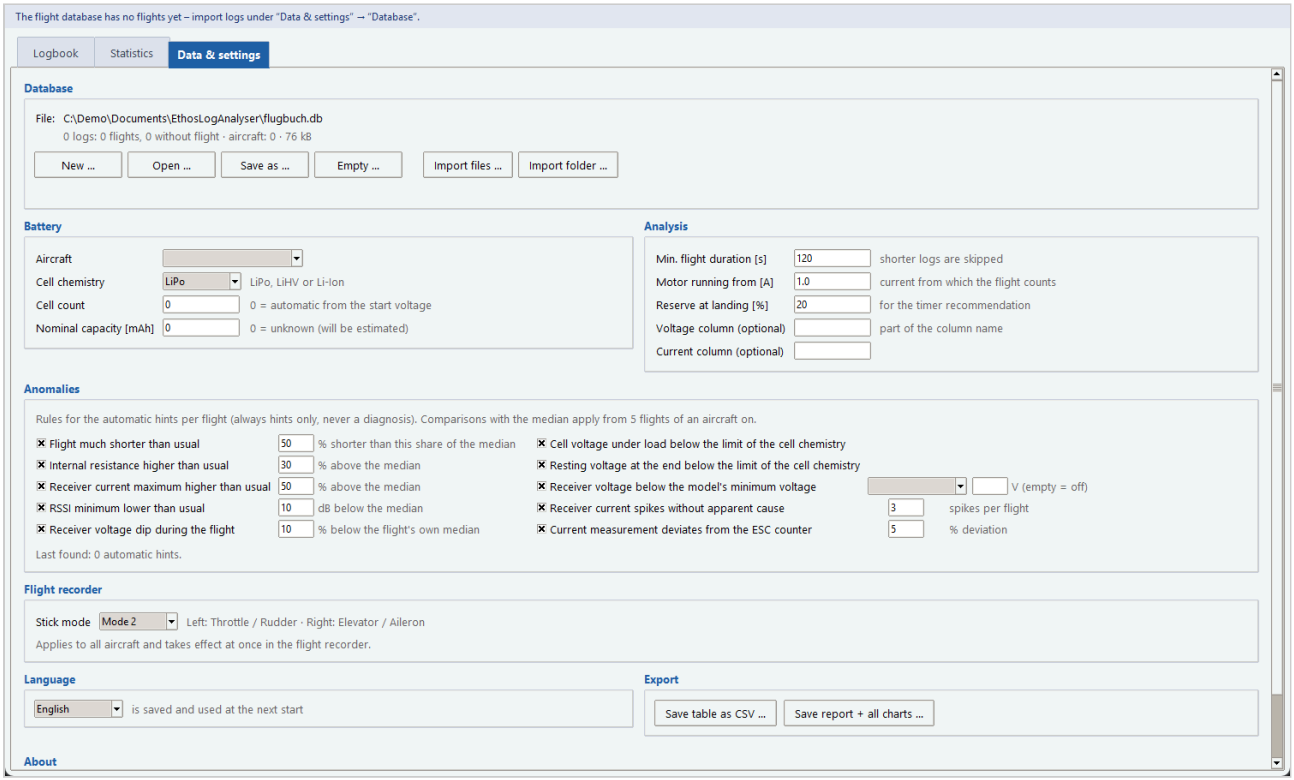
On the first start the program creates its flight database in `Documents\EthosLogAnalyser\flugbuch.db` and remembers your settings. Nothing else is installed. Please report anything that looks wrong (see Questions and problems).

2. First steps: import your logs

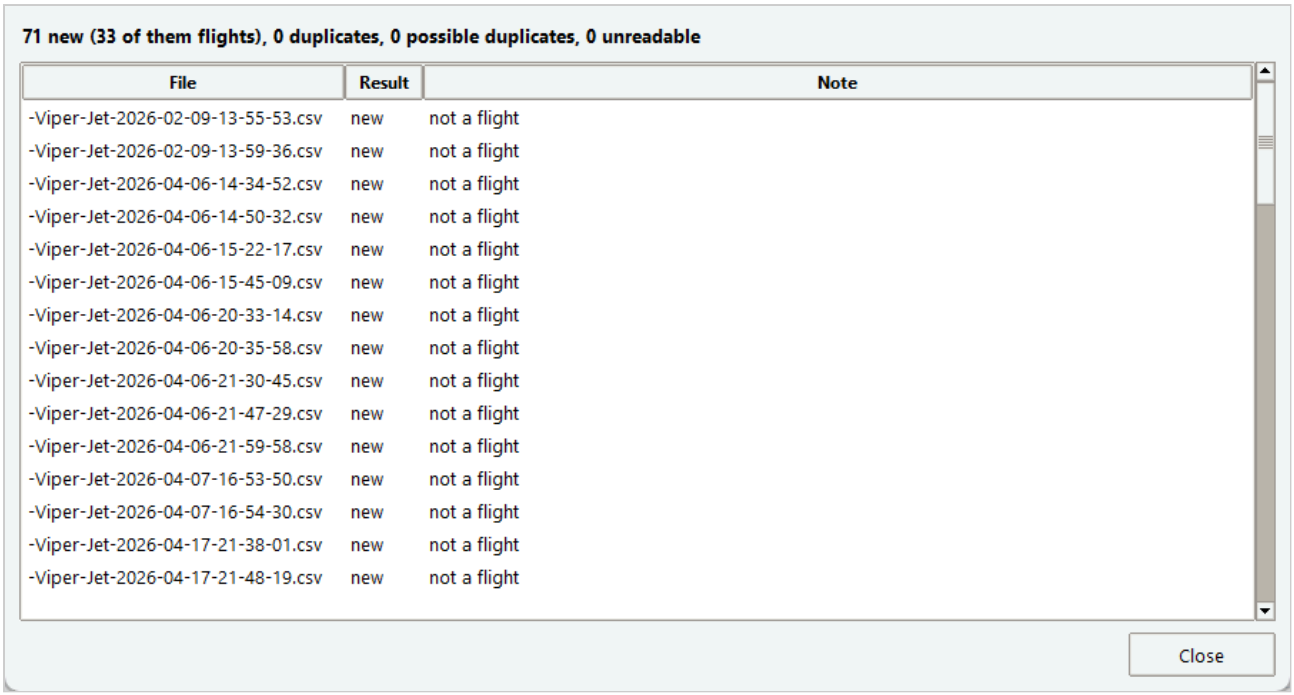
After the first start the program is empty:



1. **Get the logs.** The Ethos logs are CSV files on the SD card of your transmitter, named like `Viper-Jet-2026-04-19-13-25-36.csv`: the part before the date is the **model name**, the program uses it as the **aircraft**. Copy them to your PC, or use the SD card directly. (Logs only contain the values that are selected under *Logging* in Ethos.)
2. Open the tab **Data & settings**. The section **Database** is at the top:

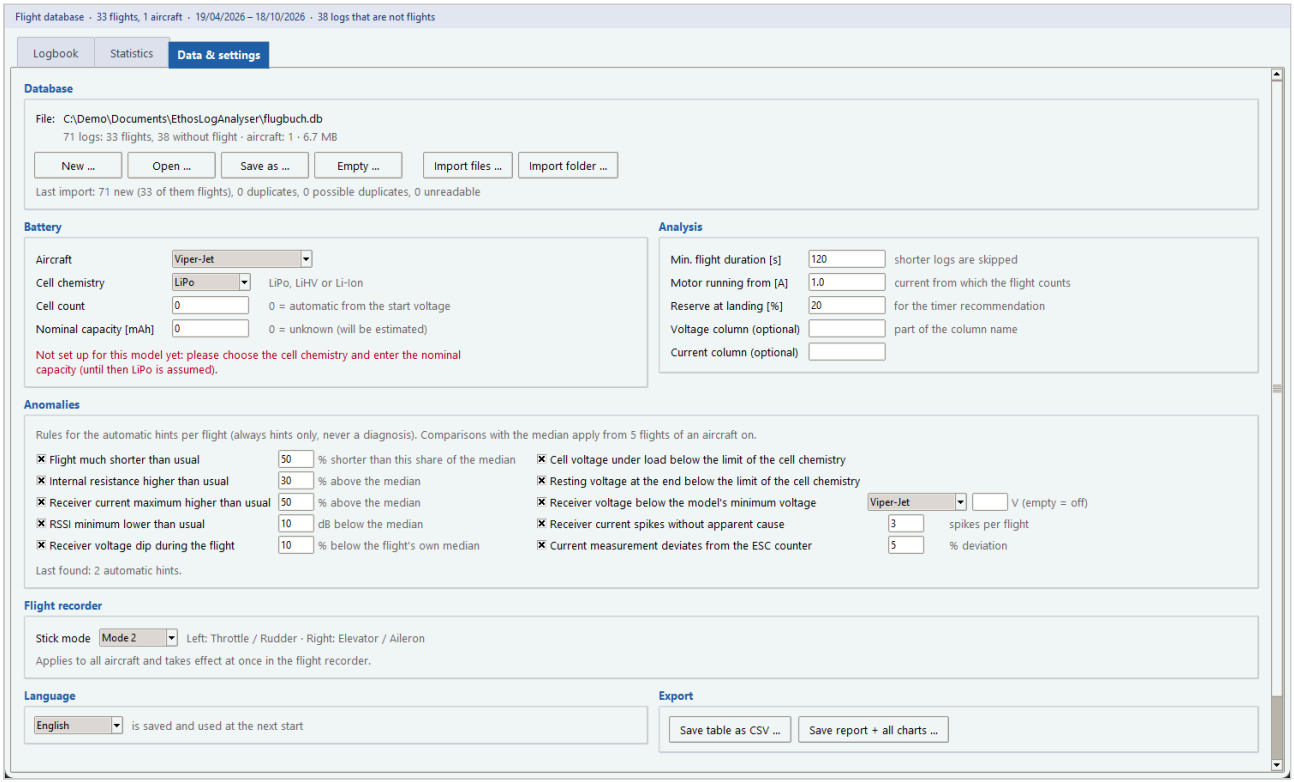


3. Click **Import folder ...** and choose the folder with your logs (subfolders are included) – or **Import files ...** for single files. A progress bar shows the import; it takes a few seconds for dozens of logs.
4. The **import report** lists every file:



Result	Meaning
new	imported
duplicate	the same content is already there (also under another file name) – skipped
possible duplicate	same aircraft and start time, other content – imported and marked with ⚠
unreadable	not an Ethos log or columns missing – the reason is shown

Every log is stored, but only real flights count: short motor tests and ground logs are marked *not a flight* (the note column says so). You can change that decision later in the logbook. 5. **Tell the program about your battery** (once per aircraft). After the import a red hint appears below the battery fields:

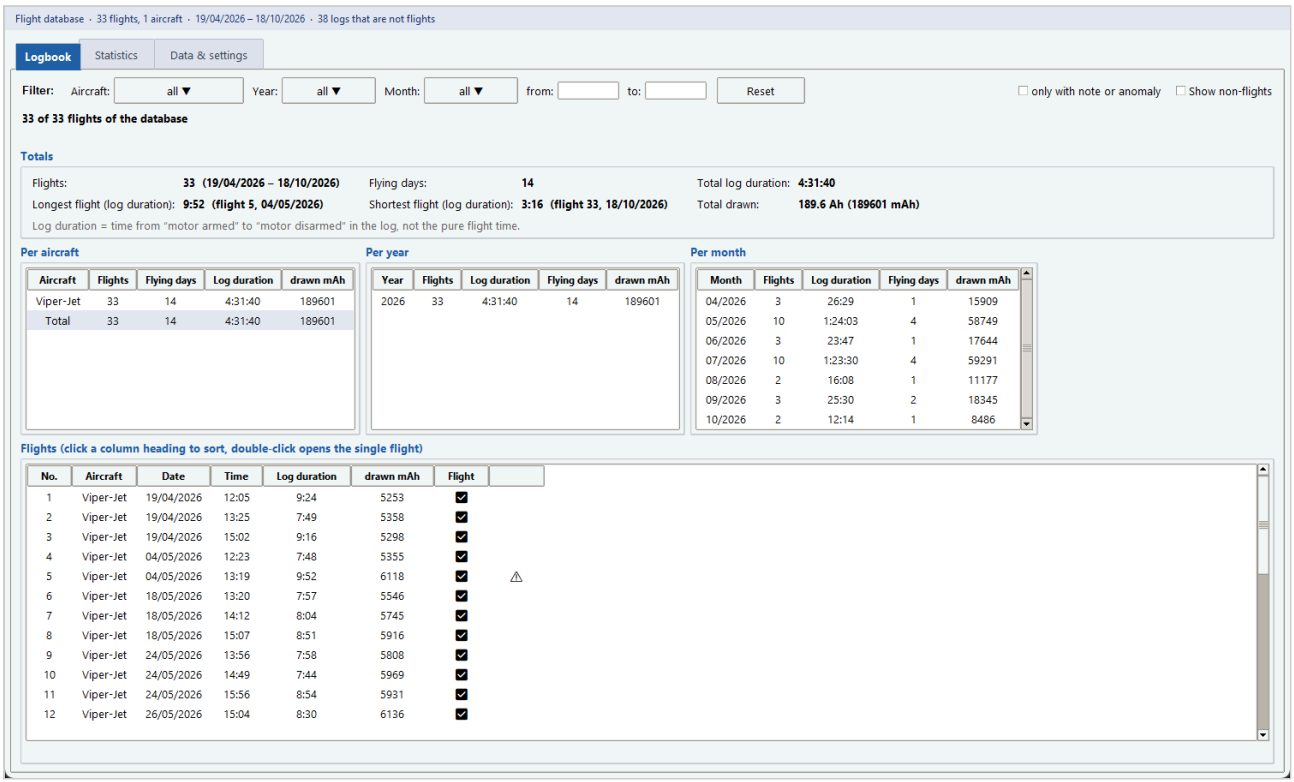


Choose the aircraft, the **cell chemistry** (LiPo, LiHV, Li-Ion) and enter the **nominal capacity** in mAh. The hint disappears. Without this the program assumes LiPo and an unknown capacity, so state of charge and battery ageing can be wrong.

You can import new logs at any time – the same file is never counted twice.

3. Logbook

The tab **Logbook** is the start page: an overview of your flying.



From top to bottom:

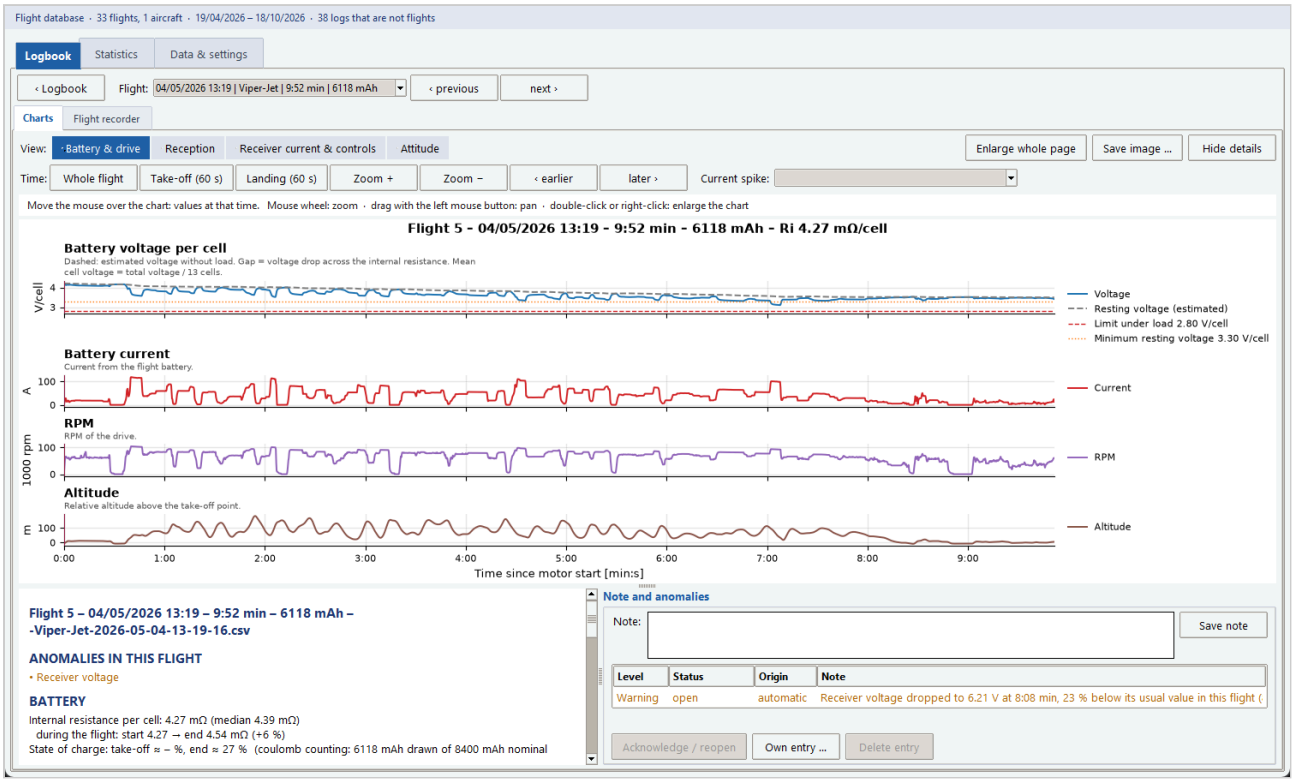
- **Filter**: show only some **aircraft**, **years** or **months**, or a date range (*from / to*, as dd.mm.yyyy). Click a field to open a list and tick several values; **Reset** removes the filter. The filter is shared with the statistics.
- **Totals**: number of flights, flying days, total log duration, longest and shortest flight, total drawn charge.
- **Per aircraft / per year / per month**: the same numbers, split up.
- **Flight list**: every flight with number, aircraft, date, time, duration and drawn mAh. Click a column heading to sort (click again to reverse). **Double-click a flight to open it** (section 4).

Good to know:

- *Log duration* is the time from “motor armed” to “motor disarmed”, not the pure airborne time. Flight numbers count per aircraft, oldest first.
- **Show non-flights** also lists logs that were not recognised as flights (grey). In the column **Flight**, a click on ☑ / ☐ changes the decision. A ☑ that you set yourself stays even after a new import.
- **only with note or anomaly** lists only flights with a note or an open hint (section 8). The last column shows ✎ for a note, **i** / ⚠ / ‼ for hints of rising severity, and ⚠ for a possible duplicate.

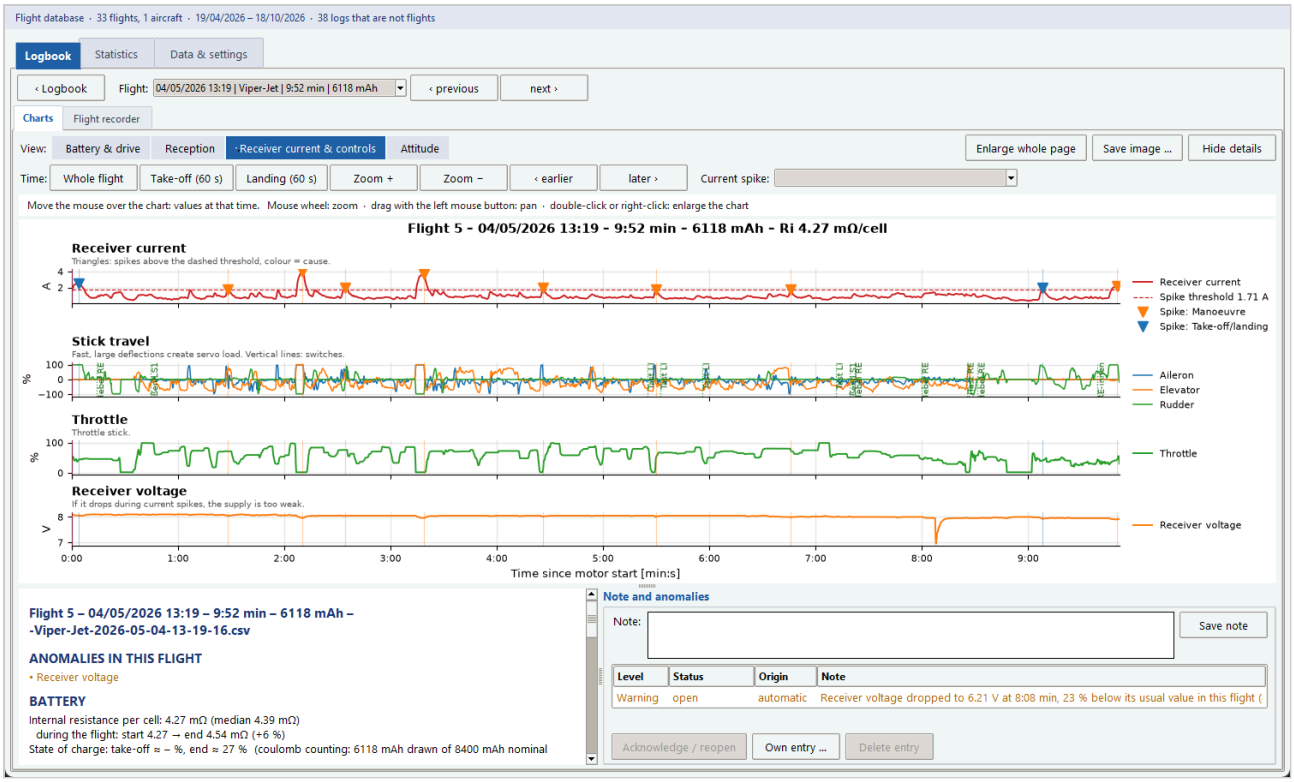
4. Single flight: charts

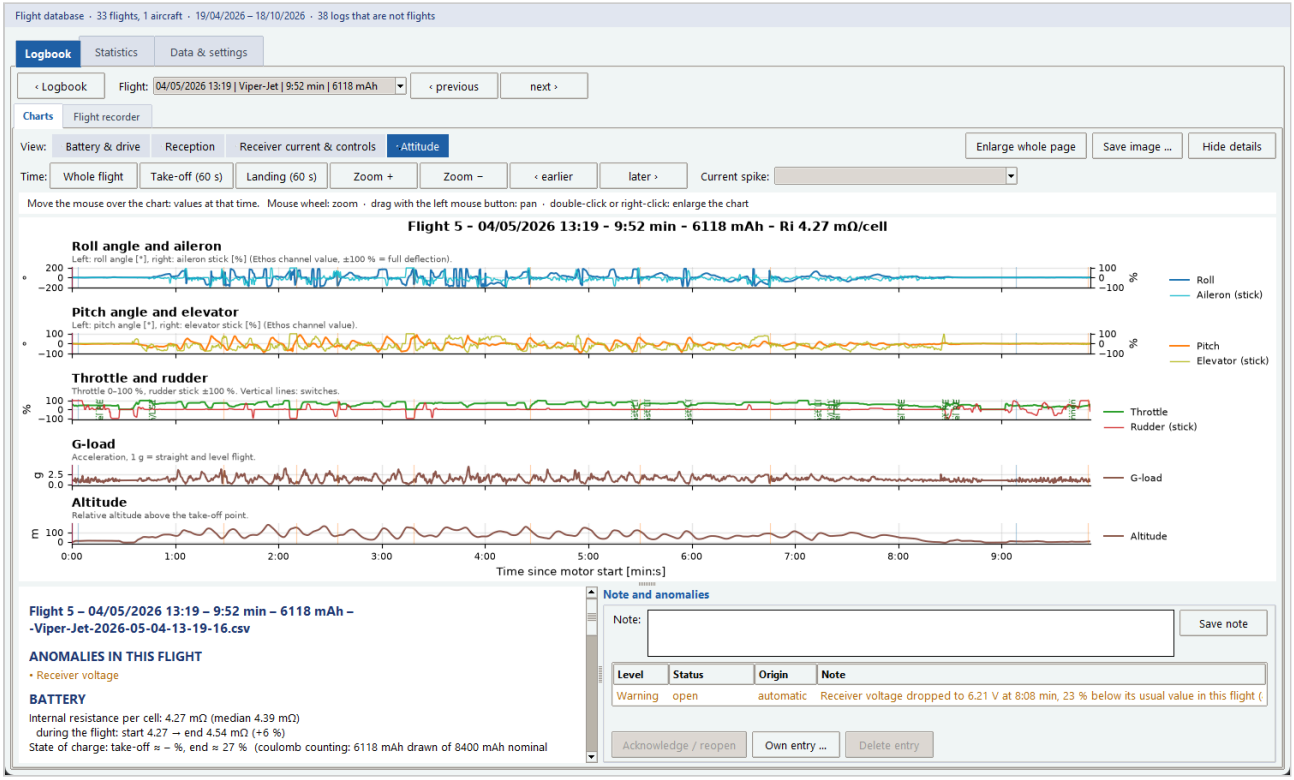
Double-click a flight in the logbook. **‹ Logbook** takes you back; **‹ previous / next ›** or the **Flight** list switch to another flight. The tab **Charts** shows the flight over time (*Time since motor start*):



Choose a **view** with the buttons above the charts:

View	Shows	Use it to
Battery & drive	voltage per cell, current, RPM, altitude	see how deep the battery sags under load
Reception	signal strength (RSSI), telemetry frames (VFR), altitude	find dropouts in the radio link
Receiver current & controls	receiver current with spikes, stick travel, throttle, receiver voltage	find out whether servo load makes the receiver supply dip
Attitude	roll, pitch, throttle and rudder with the sticks, g-load, altitude	compare the model's reaction with your stick input





Working with the charts:

- **Move the mouse** over a chart: the line above the charts shows all values at that moment, a vertical line marks the time in every chart.
- **Mouse wheel** zooms, **dragging** moves the view. **Whole flight** resets; **Take-off (60 s)** and **Landing (60 s)** jump to those parts. **Current spike** jumps to a short peak of the receiver current.
- **Double-click** a chart (or right-click → *Enlarge this chart*) shows it full screen; **Esc** closes it. **Enlarge whole page** does the same for all charts. **Save image …** saves the page as a picture.
- Below the charts the **details** compare this flight with the *median of all flights of the same aircraft* (internal resistance, state of charge, peak current and more). **Hide details** gives the charts more room.

The section *Note and anomalies* is explained in section 8.

A log without the voltage and current of the flight battery (for example a glider) has no charts, but it can still be replayed in the flight recorder.

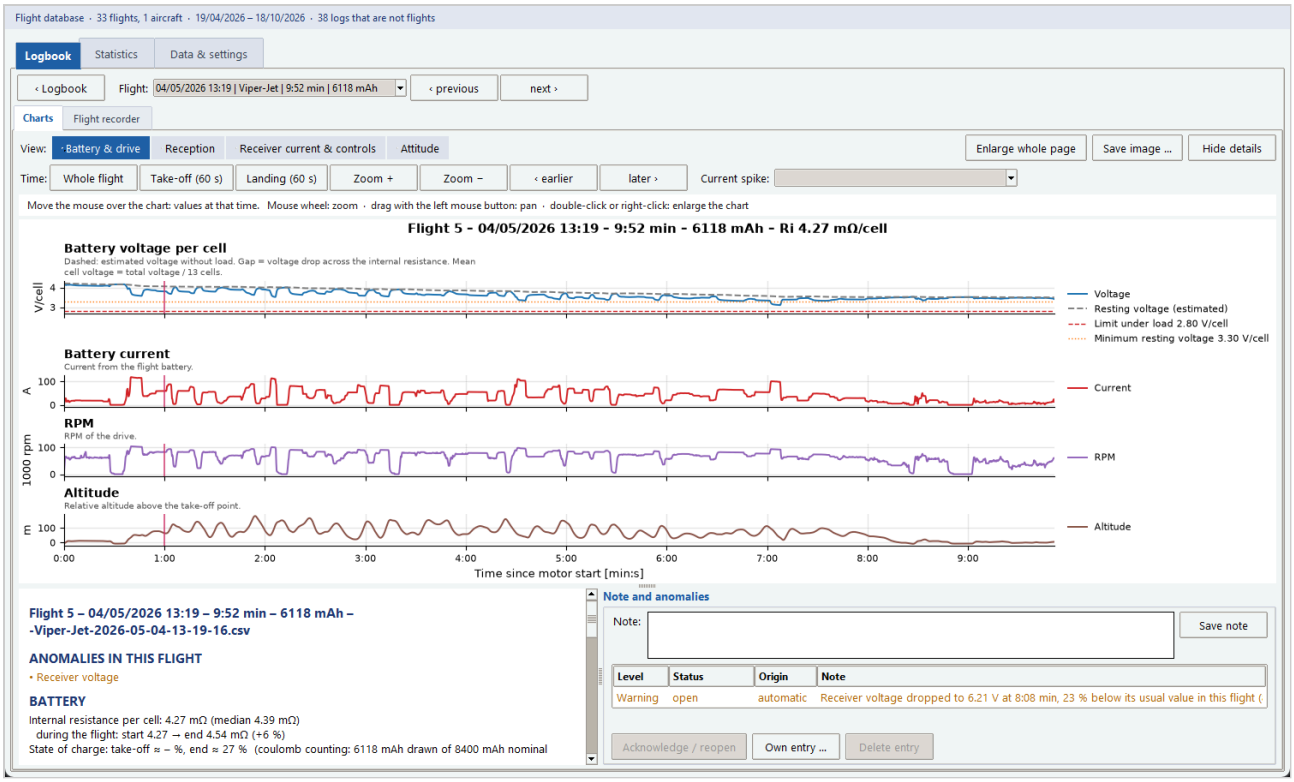
5. Flight recorder

The tab **Flight recorder** replays the selected flight in real time like a cockpit:



- **Artificial horizon** (middle): bank and pitch of the model. **Gauges**: altitude, vertical speed, drive (RPM and throttle) and battery (current and voltage). Gauges for sensors the log does not have are left out.
- **Radio** (bottom right): the two sticks as dots with the values in percent. The aircraft next to it turns with the attitude and moves its control surfaces with the sticks. It is a simple model picture, not your own aircraft.
- **Timeline** (bottom): click or drag to jump to any moment. The green band shows where telemetry exists.
- **Play / Stop**, **first / previous / next / last row**, and **Speed** from 0.25× to 16×. Keys (click into the recorder first): **Space** play/stop, ← / → one row, **Shift** + ← / → 10 s, **Home / End**.
- The log has about one row every quarter second; the picture jumps from row to row, there is no smoothing. Use 0.25× or the single steps to look at details.
- The recorder plays the same part of the flight as the charts: from motor start to motor stop.

Two things link the recorder and the charts. In the tab **Charts** a thin pink **vertical line** shows where the recorder stopped, and a **click** into a chart moves it there:



That lets you stop at an interesting moment in the recorder and read the other curves, or the other way round.

Stick mode: the sticks are drawn for the stick mode set under *Data & settings → Flight recorder* (Mode 2 is the default). Change it if your throttle sits on the other stick.

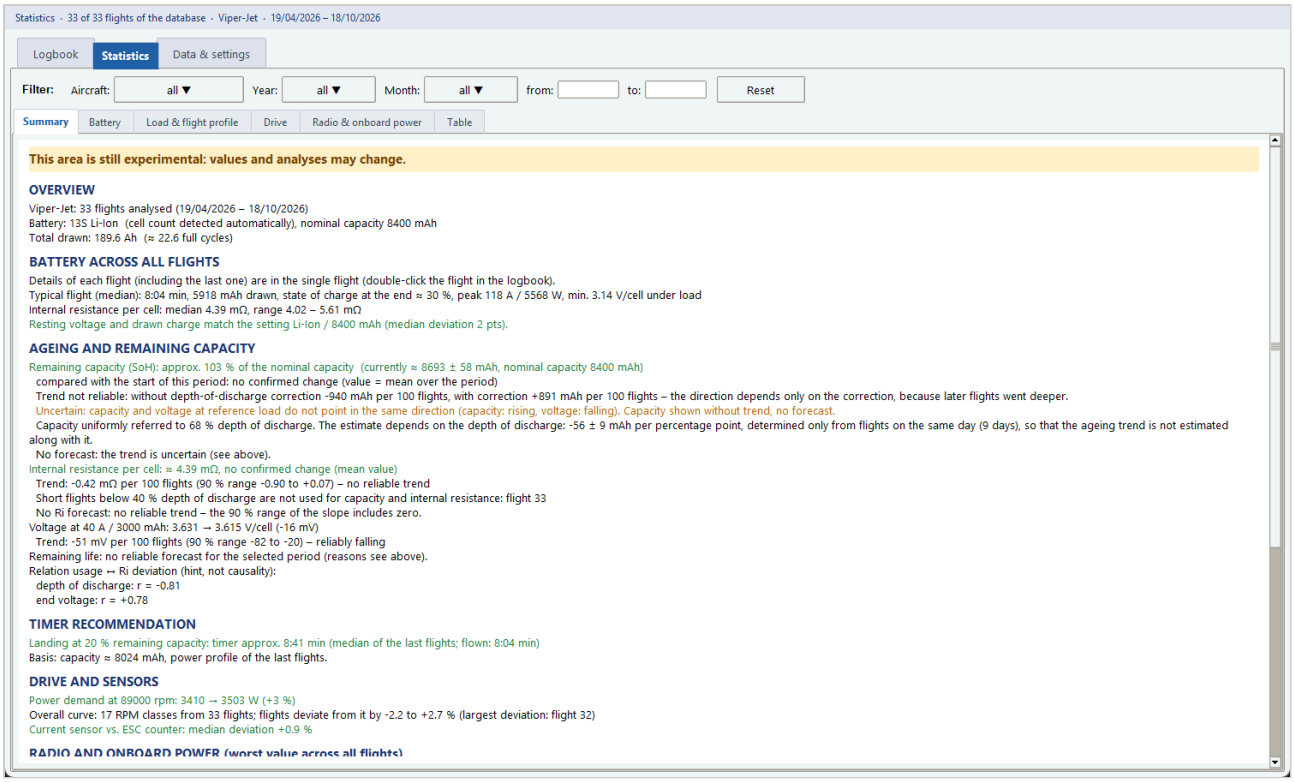
Known limitation: the direction of roll and pitch in the horizon and the aircraft has not yet been checked against all manoeuvres. If you see a model rolling the wrong way, please report it.

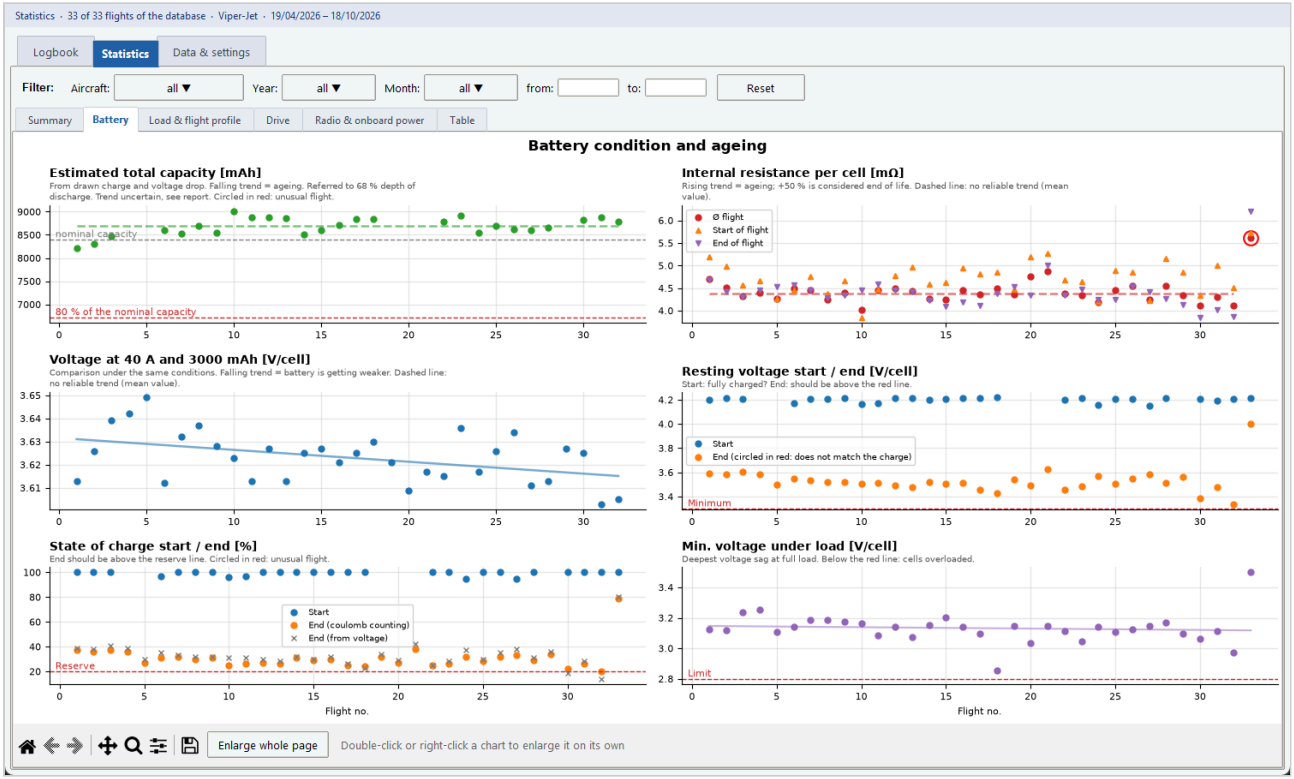
6. Statistics

The tab **Statistics** looks at **many flights of one aircraft** at once, for example how your battery ages. It has the same **filter** as the logbook. If several aircraft are selected, the program asks you to pick **one** – aircraft are never mixed. These pages are still marked *experimental*: the numbers are estimates, so treat them as hints.

The smaller tabs inside:

Page	Shows
Summary	all findings in text: battery condition, ageing and remaining capacity, timer recommendation, drive, radio and warnings
Battery	estimated capacity, internal resistance per cell, voltage at a fixed reference load, resting voltage, state of charge – one dot per flight, with trend lines
Load & flight profile	throttle histogram, C-rate, energy per flight phase, peak power, g-forces, flight time against the timer
Drive	power against RPM, power demand at a reference RPM, current sensor against the ESC counter
Radio & onboard power	RSSI, telemetry frames, receiver and transmitter battery, receiver current and its spikes, sorted by cause
Table	all key figures of every flight as a table


Statistics · 33 of 33 flights of the database · Viper-Jet · 19/04/2026 – 18/10/2026
Logbook
Statistics
Data & settings
Filter: Aircraft: all ▼ Year: all ▼ Month: all ▼ from: to: Reset
Summary
Battery
Load & flight profile
Drive
Radio & onboard power
Table
This area is still experimental: values and analyses may change.
OVERVIEW
Viper-Jet: 33 flights analysed (19/04/2026 – 18/10/2026)
Battery: 13S Li-Ion (cell count detected automatically), nominal capacity 8400 mAh
Total drawn: 189.6 Ah (≈ 22.6 full cycles)
BATTERY ACROSS ALL FLIGHTS
Details of each flight (including the last one) are in the single flight (double-click the flight in the logbook).
Typical flight (median): 8:04 min, 5918 mAh drawn, state of charge at the end ≈ 30 %, peak 118 A / 5568 W, min. 3.14 V/cell under load
Internal resistance per cell: median 4.39 mΩ, range 4.02 – 5.61 mΩ
Resting voltage and drawn charge match the setting Li-Ion / 8400 mAh (median deviation 2 pts).
AGEING AND REMAINING CAPACITY
Remaining capacity (SoH): approx. 103 % of the nominal capacity (currently ≈ 8693 ± 58 mAh, nominal capacity 8400 mAh)
compared with the start of this period: no confirmed change (value = mean over the period)
Trend not reliable: without depth-of-discharge correction -940 mAh per 100 flights, with correction +891 mAh per 100 flights – the direction depends only on the correction, because later flights went deeper.
Uncertain: capacity and voltage at reference load do not point in the same direction (capacity: rising, voltage: falling). Capacity shown without trend, no forecast.
Capacity uniformly referred to 68 % depth of discharge. The estimate depends on the depth of discharge: -56 ± 9 mAh per percentage point, determined only from flights on the same day (9 days), so that the ageing trend is not estimated along with it.
No forecast: the trend is uncertain (see above).
Internal resistance per cell: ≈ 4.39 mΩ, no confirmed change (mean value)
Trend: -0.42 mΩ per 100 flights (90 % range -0.90 to +0.07) – no reliable trend
Short flights below 40 % depth of discharge are not used for capacity and internal resistance: flight 33
No Ri forecast: no reliable trend – the 90 % range of the slope includes zero.
Voltage at 40 A / 3000 mAh: 3.631 → 3.615 V/cell (-16 mV)
Trend: -51 mV per 100 flights (90 % range -82 to -20) – reliably falling
Remaining life: no reliable forecast for the selected period (reasons see above).
Relation usage ↔ Ri deviation (hint, not causality):
depth of discharge: r = -0.81
end voltage: r = +0.78
TIMER RECOMMENDATION
Landing at 20 % remaining capacity: timer approx. 8:41 min (median of the last flights; flown: 8:04 min)
Basis: capacity ≈ 8024 mAh, power profile of the last flights.
DRIVE AND SENSORS
Power demand at 89000 rpm: 3410 → 3503 W (+3 %)
Overall curve: 17 RPM classes from 33 flights; flights deviate from it by -2.2 to +2.7 % (largest deviation: flight 32)
Current sensor vs. ESC counter: median deviation +0.9 %
RADIO AND ONBOARD POWER (worst value across all flights)

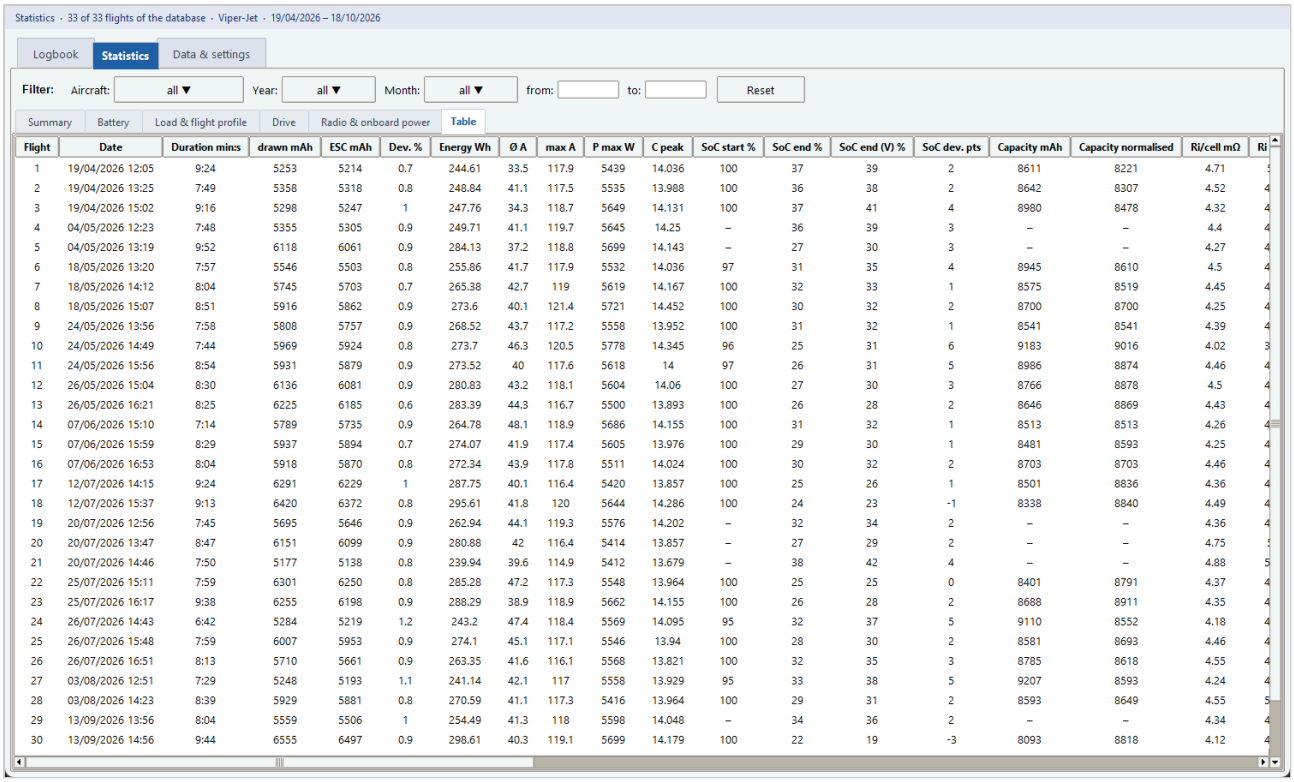

Statistics · 33 of 33 flights of the database · Viper-Jet · 19/04/2026 – 18/10/2026
Logbook
Statistics
Data & settings
Filter: Aircraft: all ▼ Year: all ▼ Month: all ▼ from: to: Reset
Summary
Battery
Load & flight profile
Drive
Radio & onboard power
Table
Battery condition and ageing
Estimated total capacity [mAh]
From drawn charge and voltage drop. Falling trend = ageing. Referred to 68 % depth of discharge. Trend uncertain, see report. Circled in red: unusual flight.
nominal capacity
80 % of the nominal capacity
Internal resistance per cell [mΩ]
Rising trend = ageing; +50 % is considered end of life. Dashed line: no reliable trend (mean value).
Ø flight
Start of flight
End of flight
Voltage at 40 A and 3000 mAh [V/cell]
Comparison under the same conditions. Falling trend = battery is getting weaker. Dashed line: no reliable trend (mean value).
Resting voltage start / end [V/cell]
Start: fully charged? End: should be above the red line.
Start
End (circled in red: does not match the charge)
Minimum
State of charge start / end [%]
End should be above the reserve line. Circled in red: unusual flight.
Start
End (coulomb counting)
End (from voltage)
Reserve
Flight no.
Min. voltage under load [V/cell]
Deepest voltage sag at full load. Below the red line: cells overloaded.
Limit
Flight no.
Enlarge whole page
Double-click or right-click a chart to enlarge it on its own



How to read them:

- **Red circles** mark unusual flights. A rising internal resistance (+50 % is considered the end of battery life) or a falling capacity points to ageing. A dashed line means “no reliable trend yet”.
- The **timer recommendation** is the flight time after which about 20 % of the capacity remains (changeable under *Data & settings → Analysis*).
- Statistics become more reliable with more flights and with the **nominal capacity** entered.
- Charts work like the charts of the single flight: hover for values, double-click to enlarge a chart, **Enlarge whole page** for all of them.

The **Table** page and the export (section 7) are meant for your own spreadsheet:

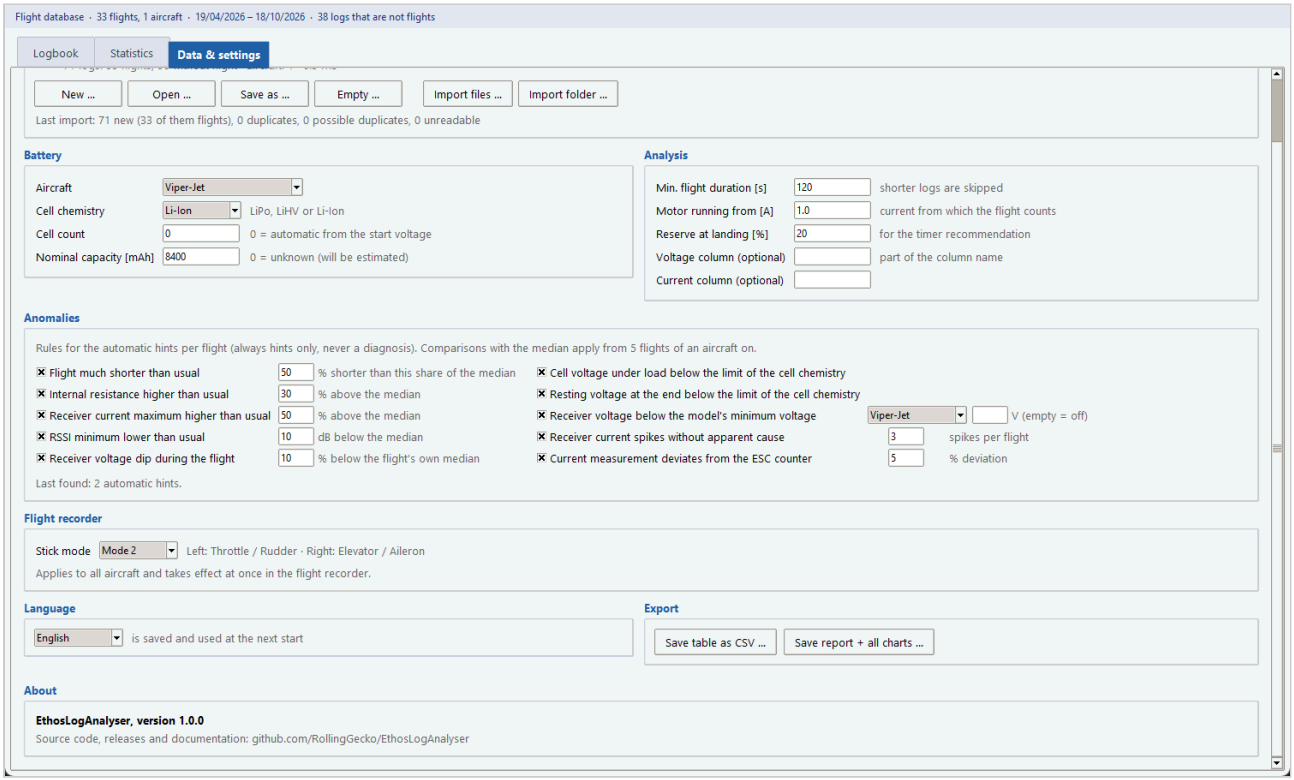


Flight	Date	Duration mins	drawn mAh	ESC mAh	Dev. %	Energy Wh	Ø A	max A	P max W	C peak	SoC start %	SoC end %	SoC end (V) %	SoC dev. pts	Capacity mAh	Capacity normalised	Ri/cell mΩ	Ri
1	19/04/2026 12:05	9:24	5253	5214	0.7	244.61	33.5	117.9	5439	14.036	100	37	39	2	8611	8221	4.71	5
2	19/04/2026 13:25	7:49	5358	5318	0.8	248.84	41.1	117.5	5535	13.988	100	36	38	2	8642	8307	4.52	4
3	19/04/2026 15:02	9:16	5298	5247	1	247.76	34.3	118.7	5649	14.131	100	37	41	4	8980	8478	4.32	4
4	04/05/2026 12:23	7:48	5355	5305	0.9	249.71	41.1	119.7	5645	14.25	–	36	39	3	–	–	4.4	4
5	04/05/2026 13:19	9:52	6118	6061	0.9	284.13	37.2	118.8	5699	14.143	–	27	30	3	–	–	4.27	4
6	18/05/2026 13:20	7:57	5546	5503	0.8	255.86	41.7	117.9	5532	14.036	97	31	35	4	8945	8610	4.5	4
7	18/05/2026 14:12	8:04	5745	5703	0.7	265.38	42.7	119	5619	14.167	100	32	33	1	8575	8519	4.45	4
8	18/05/2026 15:07	8:51	5916	5862	0.9	273.6	40.1	121.4	5721	14.452	100	30	32	2	8700	8700	4.25	4
9	24/05/2026 13:56	7:58	5808	5757	0.9	268.52	43.7	117.2	5558	13.952	100	31	32	1	8541	8541	4.39	4
10	24/05/2026 14:49	7:44	5969	5924	0.8	273.7	46.3	120.5	5778	14.345	96	25	31	6	9183	9016	4.02	3
11	24/05/2026 15:56	8:54	5931	5879	0.9	273.52	40	117.6	5618	14	97	26	31	5	8986	8874	4.46	4
12	26/05/2026 15:04	8:30	6136	6081	0.9	280.83	43.2	118.1	5604	14.06	100	27	30	3	8766	8878	4.5	4
13	26/05/2026 16:21	8:25	6225	6185	0.6	283.39	44.3	116.7	5500	13.893	100	26	28	2	8646	8869	4.43	4
14	07/06/2026 15:10	7:14	5789	5735	0.9	264.78	48.1	118.9	5686	14.155	100	31	32	1	8513	8513	4.26	4
15	07/06/2026 15:59	8:29	5937	5894	0.7	274.07	41.9	117.4	5605	13.976	100	29	30	1	8481	8593	4.25	4
16	07/06/2026 16:53	8:04	5918	5870	0.8	272.34	43.9	117.8	5511	14.024	100	30	32	2	8703	8703	4.46	4
17	12/07/2026 14:15	9:24	6291	6229	1	287.75	40.1	116.4	5420	13.857	100	25	26	1	8501	8836	4.36	4
18	12/07/2026 15:37	9:13	6420	6372	0.8	295.61	41.8	120	5644	14.286	100	24	23	-1	8338	8840	4.49	4
19	20/07/2026 12:56	7:45	5695	5646	0.9	262.94	44.1	119.3	5576	14.202	–	32	34	2	–	–	4.36	4
20	20/07/2026 13:47	8:47	6151	6099	0.9	280.88	42	116.4	5414	13.857	–	27	29	2	–	–	4.75	5
21	20/07/2026 14:46	7:50	5177	5138	0.8	239.94	39.6	114.9	5412	13.679	–	38	42	4	–	–	4.88	5
22	25/07/2026 15:11	7:59	6301	6250	0.8	285.28	47.2	117.3	5548	13.964	100	25	25	0	8401	8791	4.37	4
23	25/07/2026 16:17	9:38	6255	6198	0.9	288.29	38.9	118.9	5662	14.155	100	26	28	2	8688	8911	4.35	4
24	26/07/2026 14:43	6:42	5284	5219	1.2	243.2	47.4	118.4	5569	14.095	95	32	37	5	9110	8552	4.18	4
25	26/07/2026 15:48	7:59	6007	5953	0.9	274.1	45.1	117.1	5546	13.94	100	28	30	2	8581	8693	4.46	4
26	26/07/2026 16:51	8:13	5710	5661	0.9	263.35	41.6	116.1	5568	13.821	100	32	35	3	8785	8618	4.55	4
27	03/08/2026 12:51	7:29	5248	5193	1.1	241.14	42.1	117	5558	13.929	95	33	38	5	9207	8593	4.24	4
28	03/08/2026 14:23	8:39	5929	5881	0.8	270.59	41.1	117.3	5416	13.964	100	29	31	2	8593	8649	4.55	5
29	13/09/2026 13:56	8:04	5559	5506	1	254.49	41.3	118	5598	14.048	–	34	36	2	–	–	4.34	4
30	13/09/2026 14:56	9:44	6555	6497	0.9	298.61	40.3	119.1	5699	14.179	100	22	19	-3	8093	8818	4.12	4

Everything the program shows are **hints and estimates, not a diagnosis**. A suspicious value is a reason to look at the battery, the cables or the receiver supply – not proof of a defect.

7. Data & settings

The third tab holds everything you set up once:



Section	What it is for
Database	**New / Open / Save as / Empty** the flight database (a normal file – copy it to back up your logbook or to move to another PC) and the imports. *Empty* removes the logs from the database only; your CSV files stay untouched.
Battery	per aircraft: cell chemistry, cell count (0 = detect from the start voltage), nominal capacity (0 = unknown, will be estimated).
Analysis	*Min. flight duration* (shorter logs are not flights, default 120 s), *Motor running from* (current from which a flight counts, default 1 A), *Reserve at landing* (for the timer recommendation, default 20 %). The two column fields are only needed if the program picks the wrong voltage or current column.
Anomalies	switch the automatic hints (section 8) on or off and change their limits.
Flight recorder	stick mode 1–4.
Language	Deutsch, English, Español; the window changes at once and the choice is remembered. At the first start the system language is used if supported.
Export	**Save table as CSV** (opens in Excel; semicolon-separated) and **Save report + all charts** (text report and all charts as pictures into a folder). Both use the flights of the current statistics filter.
About	version and link to the project page.

Settings are saved automatically in `.akku_analyse.json` in your user folder.

8. Notes and anomalies

Every flight can carry a **note** and **anomalies**. Both are in the single flight, below the charts (*Note and anomalies*):

- **Note**: free text, for example "new propeller" or "hard landing". Click **Save note** (it is also saved when you leave the flight).
- **Anomalies** are hints. The program creates them by rules, for example "flight much shorter than usual", "internal resistance higher than usual", "receiver voltage dip" or "current spikes without apparent cause". Each has a level (**Info, Warning, Critical**), a status (*open / acknowledged*) and an origin (*automatic / own entry*).
- **Acknowledge / reopen** marks a hint as dealt with.
- **Own entry …** adds your own hint with a level; **Delete entry** removes your own entries (automatic ones can only be acknowledged).
- In the logbook the last column shows which flights have a note or open hints, and **only with note or anomaly** lists just those.
- The rules and limits are under *Data & settings → Anomalies*. Comparisons with the "usual" value need at least 5 flights of the aircraft.

9. Questions and problems

Problem	What to do
The logbook is empty	Import logs (section 2). Logs counted as *not a flight* are hidden: tick **Show non-flights**.
A flight is missing	Check the import report. Short logs (under 120 s) and ground logs are not flights; tick ☑ in the logbook if it was one, or lower *Min. flight duration* and import again after *Empty* ….
"Please choose an aircraft in the filter"	The statistics need exactly one aircraft: tick one in the **Aircraft** filter.

Problem	What to do
Red battery hint	Choose cell chemistry and enter the nominal capacity for that aircraft (section 2).
A flight has no charts	The log has no voltage and current of the flight battery. The flight recorder still works.
A gauge in the recorder is missing or shows “—”	The log has no such sensor, or its last value is older than 2 s.
Charts look cramped	Make the window larger. Windows display scaling of 100–150 % is supported.
Something looks wrong	Note the flight and what you did, and report it on the project page: github.com/RollingGecko/EthosLogAnalyser.

More details for the curious: Usage reference · Methodology (how the numbers are calculated) · Ethos log format.